

PHYSICS

Q.1 A ball is dropped from rest. Its speed after 2 s ($g = 10 \text{ m/s}^2$) is:

- (1) 10 m/s
- (2) 20 m/s
- (3) 30 m/s
- (4) 40 m/s

Ans. (2)

Sol. $v = gt = 20 \text{ m/s}$.

Q.2 The SI unit of force is:

- (1) Joule
- (2) Watt
- (3) Newton
- (4) Pascal

Ans. (3)

Q.3 A body of mass 2 kg moves at 3 m/s. Its kinetic energy in joules is (integer answer):

Ans. 9

CHEMISTRY

Q.1 The atomic number of carbon is:

- (1) 4
- (2) 6
- (3) 8
- (4) 12

Ans. (2)

Q.2 Which gas is produced when zinc reacts with dilute HCl?

- (1) Oxygen
- (2) Hydrogen
- (3) Chlorine
- (4) Nitrogen

Ans. (2)

Q.3 The pH of pure water at 25 C is (integer answer):

Ans. 7

MATHEMATICS

Q.1 The value of $\sin 90$ degrees is:

- (1) 0
- (2) $1/2$
- (3) 1
- (4) -1

Ans. (3)

Q.2 If $2x = 5$, the value of x is (numeric answer):

Ans. 2.5

Sol. $x = 5/2 = 2.5$

Q.3 The number of subsets of a set with 3 elements is:

- (1) 3
- (2) 6
- (3) 8
- (4) 9

Ans. (3)